Illuminator: Learning to Correct Lighting Without Ground Truth Data

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Abstract

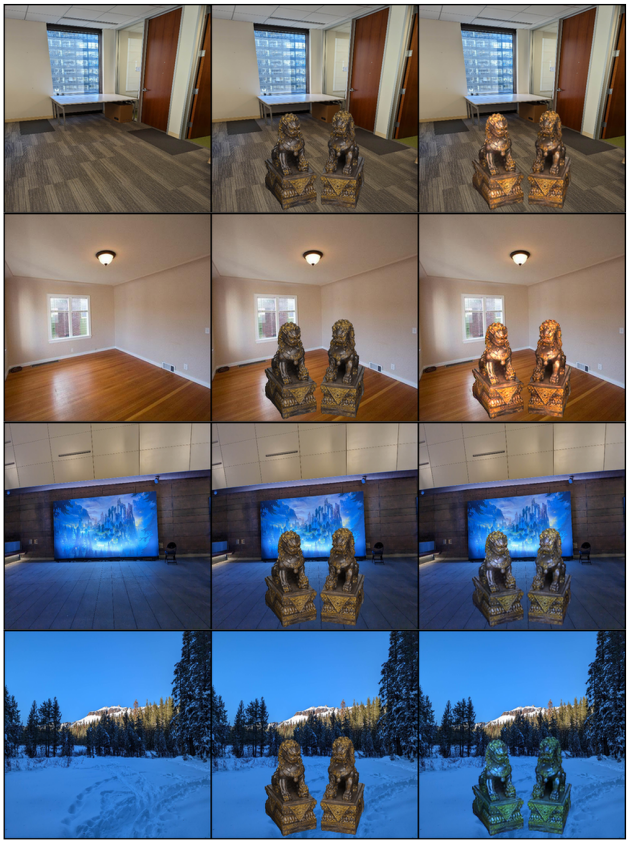
We describe a method to cut objects out of one image, move them to another, then correct the illumination to make the result look natural. Our method – Illuminator – uses unlabelled, in-the-wild images to learn what natural shading "looks like", and so can correct the unnatural shading that results from just cutting and pasting objects. Furthermore, our method can be used to change the lighting of a scene to match that of a target scene while remaining realistic.

Because it must determine what changes and what does not when an object is moved from image to image, our method must rely on an intrinsic image decomposition. Many such decompositions are available. Because we can measure the success of our method, we can choose the decomposition that produces the best results for our tasks. Our method is evaluated against several challenging baselines that depict state-of-the-art object relighting from different perspectives using a new synthetic dataset of objects inserted into multiple scenes. We show that our method outperforms all baselines in terms of visual quality and quantitative metrics. Additionally, we demonstrate the effectiveness of the learned abstraction by editing lighting in complex scenes with realistic effects. Our method is simple and easy to use, and we provide an alternative view to challenging image-based light editing tasks.

1. Introduction

Humans seem to know the difference between plausible and implausible scene lightings. For example, humans have quite strong opinions about what a scene would look like if a light was turned on. As another example, humans can spot objects that have been transferred from image to image without lighting correction. We describe a method to represent plausible lightings of a scene trained using only image data. We show how to use this method to relight images and to transfer objects from image to image. Relighting works by applying a plausible version of lighting in an example image. Transfer works by adjusting lighting on an edited image until it is plausible.

Representations of plausible scene lightings have to date been learned from massive computer graphics imagery (CGI) datasets [36, 24] or from datasets with known lighting [12, 13]. These methods compute render-oriented representations of illumination which are very good for inserting CGI objects into images but have disadvantages. First, they cannot currently adjust scene lighting in a given image toward a target illumination (as opposed to specifying appearance under particular luminaire conditions). Second, they cannot be used to transfer objects from image to image because it is currently not possible to recover a high-quality renderable CGI representation of an object like a cat or a lego



(a) Target Scene (b) Cut-and-Paste (c) Illuminator

Figure 1. Naively inserting objects into an image (a) looks wrong due to mismatching lighting between the object and the target scene (b). Our proposed method corrects the lighting mismatch and produces a plausible insertion (c), despite being only trained on image collections without geometry or lighting annotation.

block, with surface materials, from a single image. Third, they cannot currently use image data to control the effects of inference errors on the resulting rendering. Fourth, current inference procedures transfer poorly, so a method that works for indoor scenes may not work well for outdoor and vice versa [45, 21, 20].

In this paper, we sidestep CGI and instead learn lighting representations without any direct supervision and from unlabeled images, which are easy to obtain. We aggressively exploit the fact that we know how to make networks tell if the images look "right" as a result of adversarial training. Furthermore, we avoid a representation of illumination in terms of explicit physical quantities and instead allow a network to manipulate an abstract representation of scene illumination in feature space. This representation can be used both to correct the lighting of an image as well as to change the lighting of a scene in a zero-shot sense; i.e., without seeing any – labeled datasets, image pairs with and without objects, or image pairs of scenes with different lightings.

The key idea of our approach is to train a network to encode the manifold of correctly lit scenes to a set of latent variables, of which one represents an abstract representation of the scene lighting. Our design builds on top of existing, pretrained intrinsic image decomposition approaches, which decompose an image into albedo and shading images. Our network learns to encode the shading images into an abstract lighting representation, which can be easily manipulated to relight a scene. Our proposed method is orthogonal to the decomposition used and we assess the suitability of several decomposition approaches from the literature for learning a lighting representation.

To assess the efficacy of lighting representations, we select object insertion as a task for evaluation for its ability to correct inconsistent lighting when moving objects from one scene to another. As no dedicated dataset for evaluating *2D image-based* object insertion exists, we introduce a new synthetic dataset of objects inserted into multiple scenes.

In summary, our main contributions are

- We propose Illuminator, a simple and effective method for learning an abstract scene lighting representation from in-the-wild images.
- We introduce a novel cross-illumination cycle consistency loss, which allows training Illuminator without any paired images or labeled illumination datasets.
- We demonstrate how the learned lighting representation corrects the appearance of objects when transferring them between scenes (as 2D image fragments, and not 3D objects).
- We introduce a challenging new synthetic dataset for evaluating object relighting.
- We analyze state-of-the-art image-decomposition methods for learning scene lighting abstraction, discuss their limitations, and propose improvements.
- We further demonstrate the versatility of Illuminator and its learned light representation by editing the lighting of complex scenes.

2. Related work

Lighting representation. Many image synthesis and manipulation tasks, such as object insertion and relighting, require establishing the illumination from the image observation. Lighting is typically represented in a parametric model suitable for rendering. Such models are typically environment maps (equiv. panoramas) or parametric luminaires. Debevec [8] recovered an environment map from images of mirrored spheres in the environment. Karsch *et al* infer visible luminaires [16]; in later work, they showed an entire environment map could be imputed by matching [17]. Gardner *et al.* use a learned method to recover a panorama representing illumination, which is warped to correct for camera position [12]; improvements follow for indoor scenes by recovering a fixed size set of virtual parametrized luminaires [11]. The spherical harmonic expansion of [35] is now widely adopted. Li et al. [20, 21] represent the illumination field as a dense 2D grid of spherical harmonic sources. Similarly, Wang et al. [44] learn a 3D volume of spherical Gaussians. While these methods can model complex light-dependent effects, they require large CGI datasets [36, 24] and dataset of illumination maps [12, 13] for training. In contrast, our approach uses no labeled or CGI data and produces an abstract representation of illumination as a set of deep features without any specific physical interpretation.

Object model insertion uses an imputed illumination field and a detailed computer graphics model to shade that model and insert it into the scene. Karsch et al. [16, 17], Kholgade et al.,[18] and [21] show strong results. But CGI insertion methods require acceptable illumination and accurate object models with relightable materials. One can obtain these models from sufficient images – for example, [6] insert cars from a collection of 3D reconstruction assets into scenes – but inferring an object model from a single real-world image remains hard at present [23, 2].

Image fragment insertion takes an object from one image and moves it to another. The approach was pioneered by [19], but cannot correct lighting – so, for example, a fragment cut from a bright image and placed in a dark image will seem to glow; careful matching of an object to a scene can help [43]. Liao et al., [25, 26] can correct long-scale shading effects for inserted fragments using a physical model. In contrast, our method learns to automatically adjust the lighting to produce an image that is like real images. Bhattad and Forsyth [3] adjust the shading field using a deep image prior [42] with surface normal supervision. In contrast, our method requires no geometric supervision. Image harmonization (IH) methods [40, 41, 7, 28] could be used to produce a rendered image, but tend to adjust the color appearance of the object

rather than appearance due to lighting changes. In contrast, our method can correct a fragment without misrepresenting its color or any other intrinsic properties.

Image-based relighting. Recent works show portrait images can be convincingly relighted [39, 47, 33, 37]; these methods are specialized to faces, and are trained using paired or light-stage data. In contrast, we show acceptable complex scene relighting results from our abstracted illumination fields. Another line of work exploits a self-supervised method for outdoor image relighting. Such approaches leverage single-image decomposition with parametric outdoor illumination, which is more straightforward than indoors as they are often dominated by sky and sunlight [46, 29]. In contrast, our illumination field aims to handle a more complicated spatially-varying illumination field suitable for both indoor and outdoor scenes.

Decomposition of an image into albedo and shading is key to our method, and is rooted in intrinsic image decomposition [22, 4, 14, 9]. A review of these methods is in [10]. We study different state-of-the-art methods for learning lighting representations and show that the choice of decomposition method is crucial.

Cycle consistency. To force our method into learning an abstract lighting representation, we exploit a cycle consistency loss. Such cycle consistencies have been used in various computer vision tasks [48, 1, 31, 32, 49, 2]. Most of these works deploy cycle consistency to improve the perceptual quality of rendered images. Yu et al[46] use a cycle consistency for learning to relight outdoor scenes. While they apply the cycle consistency to aid learning a more stable decomposition into geometry, materials, and lighting and leverages multiple views of the same scene under different lighting conditions for training, our approach makes fewer assumptions about the scene decomposition and does not require paired images for supervision.

3. Overview

We provide an overview of our approach in Figure 2. Our method builds on top of existing intrinsic image decomposition approaches, which decompose an input image into an albedo and a shading image. The albedo image encodes the surface albedo of all visible surfaces in an image. Intuitively, the albedo images represent the "color" of objects if all lights were removed from the scene, and the shading image encodes the effects of the scene lighting when all objects were to be white. Consequently, the shading image visualizes interactions of the light with the scene geometry.

Illuminator operates on the albedo and shading images and encodes them into latent representations that we term persistent latent variable (PLV) and transient latent variable (PLV). Decoding the latent variables again renders an image. Illuminator's network design is thus straightforward and resembles the fairly standard encoder/decoder approach taken by many approaches to various perception tasks.

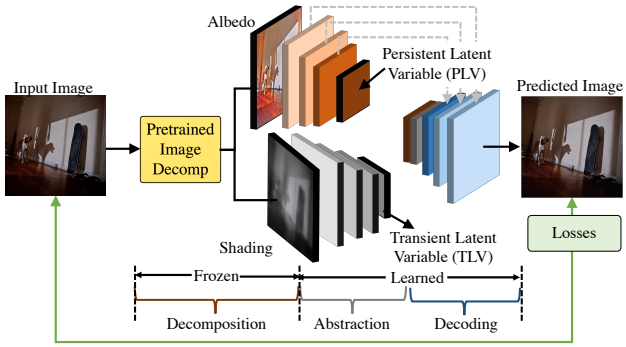


Figure 2. **Overview**: Illuminator builds on top of pretrained image decomposition approaches. It uses their potentially imperfect decomposition into albedo and shading to learn an abstract lighting representation. To that end, it encodes albedo into a persistent latent variable, and shading into a transient latent variable (TLV). Decoding the latent variables renders an image. We force TLV to represent an abstract lighting representation by training Illuminator with a novel cross-illumination consistency loss (Fig. 3).

The key idea to learning an abstract and versatile lighting representation comes from the careful design of the training objective, which we detail in Section 3.2. While the design of our network and the training objective are compatible with a wide variety of existing image decomposition approaches, the particular choice significantly affects the representation learned, and thus the quality of images produced in downstream applications. We assess the suitability of image decomposition approaches to learning a high-quality lighting representation through object insertion as a downstream task. To that end, we collect a new dataset without ground truth for inserting and relighting objects into scenes.

3.1. Network

Illuminator consists of a pretrained image decomposition approach Ψ, an encoder network E, and a decoder network D. Feeding an image I to Illuminator, it is decomposed into components $\Psi_j(I)$, with the number of components depending on the approach chosen (e.g. $j = \{1, 2\}$ for albedo & shading). The components are encoded into latent variables π (persistent) and τ (transient):

$$(\phi, \tau) = E\left(\Psi_*(I)\right).$$

Decoding the latent variables reconstructs an image $I' = D(\phi, \tau)$.

3.2. Training objective

We train Illuminator through a reconstruction objective, which ensures that encoding an image into latent variables and decoding them again produces an image equal to the input. We further propose a cross-illumination cycle consistency loss, which is the key to learning an abstract scene lighting representation from unlabeled image collections.

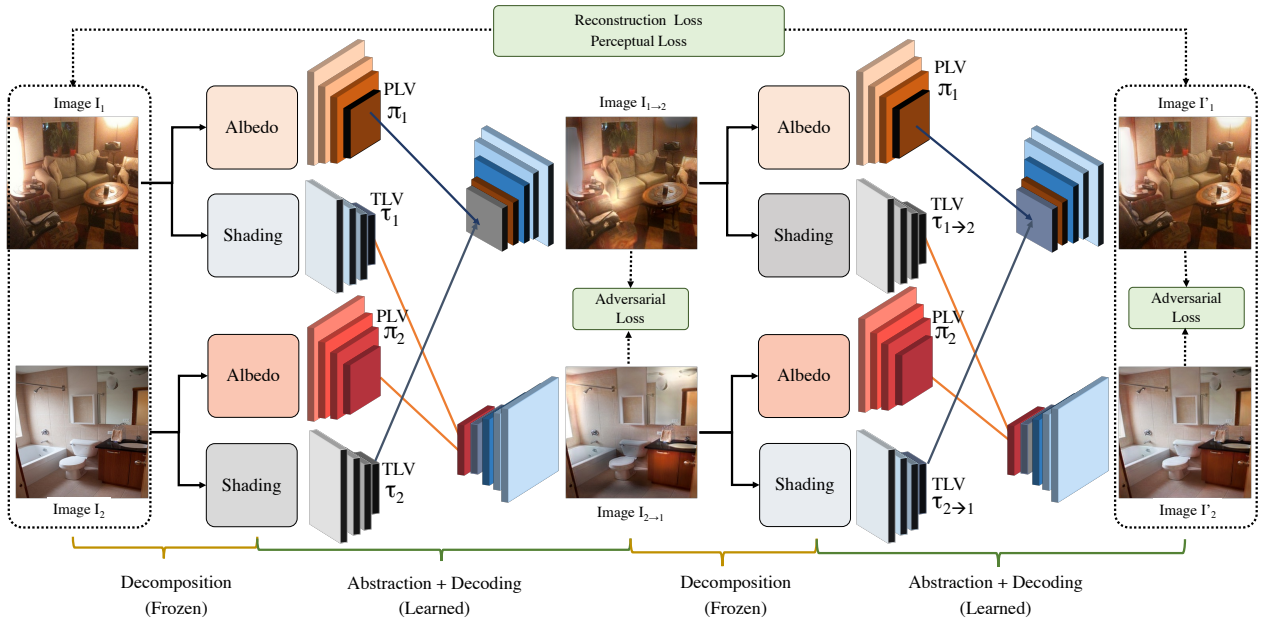


Figure 3. During training, we decompose images I_1 & I_2 into persistent and transient latent variables, which we mix across images to obtain relighted images $I_{1\to2}$ & $I_{2\to1}$. Applying the same procedure to $I_{1\to2}$ & $I_{2\to1}$ yields images $I_1^{'}$ & $I_2^{'}$, which should equal the original I_1 & I_2, establishing a **cross-illumination cycle consistency.** We enforce consistency through typical reconstruction & perceptual losses, see text for details. Decoding images from latent variables is supervised through an adversarial loss. Notice how light sources in the scenes are swapped after the first half of the cycle ($I_{1\to2}$ & $I_{2\to1}$).

Reconstruction objective. To ensure that the reconstructed image matches the input image, we enforce a reconstruction objective, consisting of several standard objectives. Specifically, we apply an $\mathcal{L}_1$ (reconstruction) loss

$$\mathcal{L}_{rec} = ||I - I'||_1,$$

and a perceptual loss [15] which operates on feature maps extracted at layers Φ_j of a pretrained VGG network [38] to ensure that I' matches I directly.

$$\mathcal{L}_{per} = \sum_j ||\Phi_j(I) - \Phi_j(I')||_2^2,$$

We further leverage the chosen decomposition approach for additional consistency in the space of decompositions:

$$\mathcal{L}_{decom} = \sum_{j=1}^{N} ||\Psi_j(I_i) - \Psi_j(I_i')||_2.$$

Finally, we penalize reconstructions that look unrealistic through a standard non-saturating adversarial loss $\mathcal{L}_{adv}$ (details in the supplement).

The reconstruction objective thus becomes

$$\mathcal{L} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{per}\mathcal{L}_{per} + \lambda_{decom}\mathcal{L}_{decom} + \lambda_{adv}\mathcal{L}_{adv},$$

with weights λ_* specified in the supplement.

Cross-illumination cycle consistency. Simply passing intrinsic image decompositions through an encoder does not guarantee that any of the latent variables represent a meaningful lighting representation and that it is decoupled from a scene representation. Starting with the decompositions, we have found that all current approaches to intrinsic image decomposition in practice leave traces of the original image's shading in the albedo map. But more importantly, even a correctly estimated shading map would by definition entangle the geometry and lighting of a scene.

Shading at a point is determined by both the normal at that point and the surface irradiance. For example, the shading field of a cluttered room and the same room without clutter might be wildly different, even if a sconce on the wall illuminates both. To ensure that the TLV encodes only lighting and not scene, we need it to encode the distribution of irradiance but not the surface normals. In the example, it should encode the presence of the sconce without the normals.

To disentangle geometry and lighting, and learn an abstract lighting representation from given image decompositions, we propose a cross-illumination cycle consistency, visualized in Figure 3.

We randomly select two images I_1, I_2 from the training set and let the pretrained decomposition approach decompose them into albedo and shading images. Illuminator then encodes each albedo image into a persistent latent variable $\pi_{\{1,2\}}$ and each shading image into a transient latent variable $\tau_{\{1,2\}}$. We then mix the latent variables and feed them to the

decoder D to reconstruct intermediate images $I_{1\to2}, I_{2\to1}$:

$$I_{1\to2} = D(\pi_1, \tau_2)$$
$$I_{2\to1} = D(\pi_2, \tau_1).$$

Applying the same procedure (intrinsic image decomposition, encoding, mixing, decoding) to the intermediate images, we obtain reconstructions I'_1, I'_2 of the original images.

$$I'_1 = D(\pi_1, \tau_{2\to1})$$
$$I'_2 = D(\pi_2, \tau_{1\to2}).$$

We apply the reconstruction objective to the pairs I_i, I'_i as previously, and further constrain the intrinsic image decompositions of the intermediate images:

$$\mathcal{L}_a = ||\Psi_{albedo}(I_{1\to2}) - \Psi_{albedo}(I_1)||_2 + ||\Psi_{albedo}(I_{2\to1}) - \Psi_{albedo}(I_2)||_2$$

To ensure a realistic appearance of reconstructed images, we again apply an adversarial loss.

3.3. Variant decomposition

The choice of the intrinsic image decomposition is crucial for learning an abstract lighting representation as we demonstrate through a controlled experiment in Sec. 4.3. For Illuminator, we extend the self-supervised Paradigms by Forsyth & Rock [10]. Forsyth & Rock train intrinsic image methods using samples from abstract statistical models (paradigms), where albedo paradigms are Mondrian-like images and shading paradigms are constructed out of Perlin noise [34]. Their method trains a U-net to decompose simulated images made by multiplying samples drawn from a statistical model of albedo with samples drawn from a statistical model of shading. An adversarial loss encourages decompositions of real images to "look like" albedo and shading samples. A consistency loss ensures decompositions of real images composed to produce the original image. No CGI data is required.

We exploit the fact that their model is extensible; in particular, while they decompose images as image = albedo × shading, we decompose as image = albedo × shading + gloss. We use versions of their models for albedo and shading, and smooth Perlin noise for gloss. Samples of our albedo, shading and gloss are in Figure 4.

We train our approach on images from MSCOCO [27] without using any of its labels. We pick MSCOCO because it contains a wide variety of objects and lighting scenarios, featuring indoor and outdoor scenes. For further details about the network architecture, optimizer, and training schedule we refer to the supplement.

4. Experiments

We evaluate Illuminator's ability to learn an abstract lighting representation through a downstream task. Specifically,

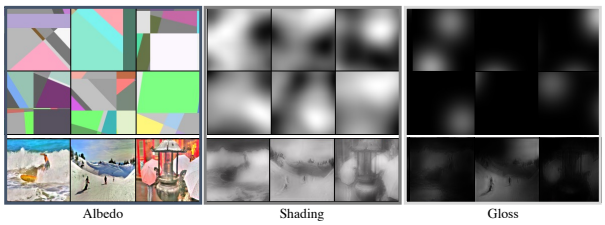


Figure 4. Top two rows: samples of our variant decomposition that we use to train our image decomposition model. Bottom row: decomposition from the randomly chosen MSCOCO images. Unlike traditional albedo and shading decomposition, our variant decomposition preserves most of the geometric and high-frequency details in the albedo map and hence enables us to transfer objects without loss of surface details.

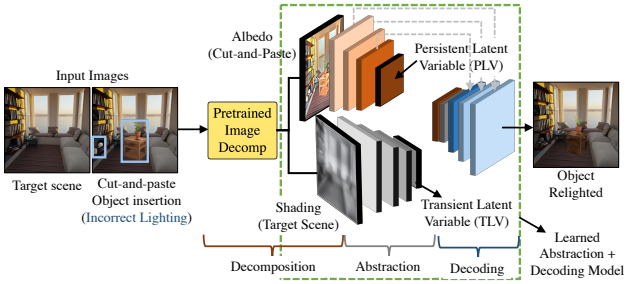


Figure 5. **Illuminator can correct an object's inconsistent lighting at test time.** After training an abstraction + decoding model from intrinsic images, at test time, Illuminator can correct the lighting of a naive cut-and-paste insertion. We decompose the target scene and cut-and-paste inserted scene together. We extract PLV from the cut-and-paste image and TLV from the target scene before inserting objects. We observe Illuminator corrects inconsistent lighting on the inserted fragment; having seen no labeled paired data of scenes with and without objects. This is akin to a zero-shot lighting correction setting.

we insert objects into images, resulting in a mismatching appearance of the objects. We then use Illuminator to correct the lighting of the object. As no suitable dataset for 2D object relighting exists, we collect one.

4.1. Dataset for object relighting.

We rendered 12 CAD models with complex material and surface properties in nine different environments (including indoors and outdoors) with two to three different lighting conditions for each. From these renders we obtain 4752 quantitative comparisons for object insertion. We provide details in supplementary. To facilitate further research, we will make the evaluation dataset publicly available.

4.2. Comparison to prior work

We compare our work against several baselines, representative of different views on object insertion.

Naive baselines. As a lower bound we use a naive cut-and-paste object insertion, i.e. the lighting of the inserted object is not modified at all. We term this baseline *cut-and-paste*.

Table 1. Results for relighting inserted objects. We compare against 2 naive baselines, 2 state-of-the-art approaches to image harmonization, and an oracle with access to ground truth geometry. Our method outperforms all baselines except the oracle. Removing the proposed cross-illumination cycle-consistency loss (*w/o CICC*) results in a worse performance as measured by PSNR.

Method	Training labels	PSNR↑	LPIPS↓
Cut-and-Paste	✗	15.08	0.042
Oracle (based on [20])	✓	18.84	0.033
Dovenet [7]	✓	16.61	0.048
Rainnet [28]	✓	16.23	0.050
Match-intensity	✗	16.24	0.040
Ours (w/o CICC)	✗	17.16	**0.038**
Ours	✗	**17.65**	**0.038**

As another naive baseline, we simply correct the inserted object's intensity by decomposing it into intrinsic image components, adjusting the albedo estimate of the object to match the albedo of the target scene, and recompose the image components. This effectively matches the object's appearance to the color of the target illumination, assuming a white illumination in the source image (cf [5]). We term this baseline *match-intensity*.

Image harmonization baselines. We further compare against two state-of-the-art approaches to image harmonization, namely Dovenet [7] and Rainnet [28]. Different from Illuminator, both approaches are trained on labeled data. Furthermore, they are both trained to match the color intensity rather than adjust the lighting.

As an upper baseline, we render the ground truth 3D objects (*not* fragments) using environment maps predicted by [20]; as this baseline (termed *Oracle*) has access to ground truth geometry and materials, we expect it to provide an upper bound. We present quantitative results on our dataset in Table 1, and qualitative samples in Figure 6.

As expected, the naive cut-and-paste baseline performs worst in all metrics. Surprisingly, the simple match-intensity baseline performs on par with the image harmonization baselines. Our approach outperforms all baselines except the oracle with access to ground truth geometry. Removing the cross-illumination cycle-consistency loss reduces the performance of our method as measured by PSNR, indicating that the proposed loss helps to learn a better representation.

4.3. Choice of image decomposition

To assess how our intrinsic image decomposition approach affects the quality of Illuminator's learned lighting representations, we compare it against several baselines, which represent the state of the art for different views on intrinsic image decomposition; (1) *CGIntrinsics* [22] is a feedforward network that directly predicts shading and reflectance. (2) *SVBRDF* [20] is an inverse rendering approach that predicts dense spatially varying BRDFs. (3) *Paradigms* [10] is a self-supervised approach that predicts albedo and shading. (4) Our approach builds on Paradigms (as detailed in Section 3.3).

Table 2. **Choice of image decomposition.** We compare several state-of-the-art approaches to intrinsic image decomposition to produce the decompositions, which we feed into Illuminator's encoder. For all approaches, we train Illuminator on the decompositions and report results on our dataset. We find that CGIntrinsics performs better than SVBRDF and Paradigms, and our variant decomposition (Ours) outperforms all evaluated baselines.

Method	labeled data	PSNR↑	LPIPS↓
CGIntrinsics [22]	✓	15.68	0.046
SVBRDF [20]	✓	14.60	0.050
Paradigms [10]	✗	15.20	0.046
Ours	✗	**17.65**	**0.038**

Table 3. WHDR evaluated on IIW dataset for different image decomposition methods.

CGIntrinsics [22]	SVBRDF [20]	Paradigms [10]	Ours
14.80	16.83	16.86	19.00

We again evaluate on our dataset. Results are shown in Table 2 and qualitative samples in Figures 6 & 7.

WHDR does not predict relighting score: From Table 2, SOTA intrinsic image methods produce relatively weak results. Our variant decomposition achieves a relatively weak WHDR score of 19 (compare SOTA CGIntrinsics [22] of 14.8, and Table 3), but produces very strong relighting results. Figure 6 and Figure 7 suggest the cause — the image decompositions we have used do not preserve properties that have important effects on the rendering. For example, no current method can recover the geometry of a stud on a lego brick from a single image, meaning that the small shadow cast by the stud should be modeled as persistent – because then it will follow the brick when it is moved from the image to image. In contrast, our variant decomposition produces strong relighting results (Figure 7 and Figure 8).

4.4. Further applications

We apply Illuminator to relighting of a complete scene in Figure 9). To edit the scene lighting, we manually edit the gloss map produced by our intrinsic image decomposition approach and pass it to the rest of the pipeline.

5. Discussion and Conclusion

Illuminator offers a simple and effective solution for correcting inconsistent lighting in inserted 2D image fragments without the need for a 3D CGI model or illumination maps during training. The method is based on a pretrained intrinsic decomposition model trained on statistical models of Retinex, which outperforms several state-of-the-art intrinsic image decomposition techniques that rely on labeled CGI datasets. Illuminator is able to achieve comparable performance to SVBRDF [20], which inserts 3D CGI objects (upper baseline). Illuminator's ability to correct lighting in

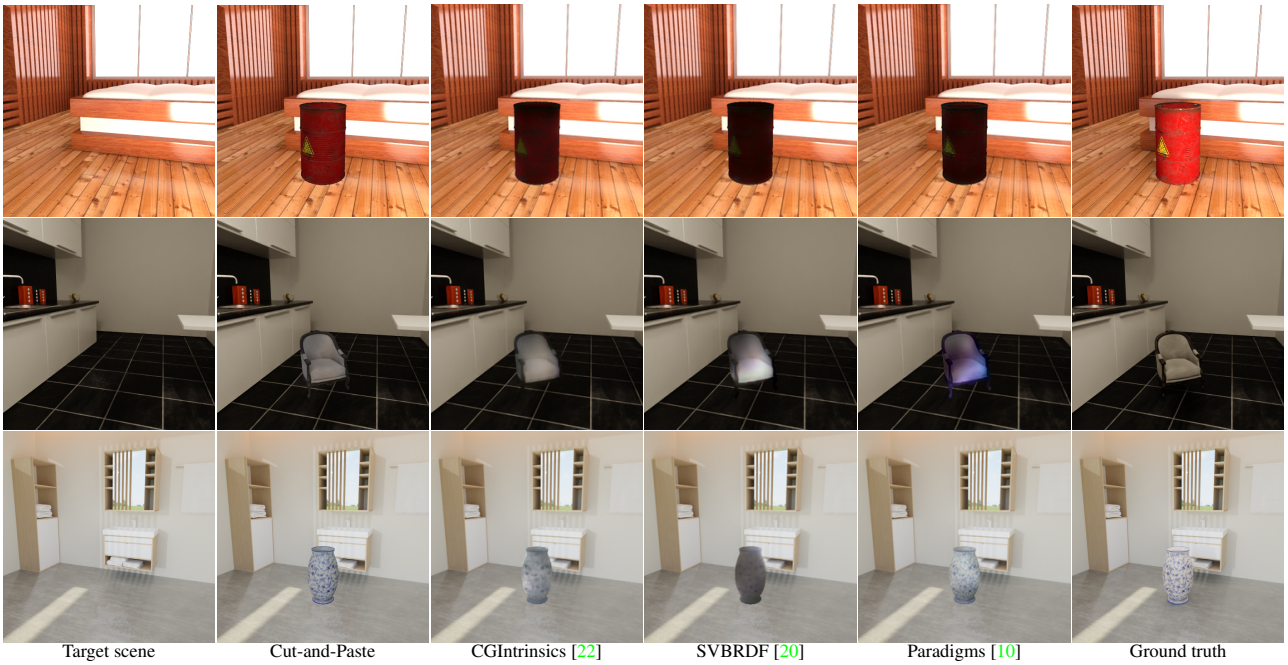


Figure 6. We employ several state-of-the-art decompositions to assess their suitability for relighting an inserted object. For all tested approaches, the relighted object consistently differs significantly from the ground truth (right-most column).

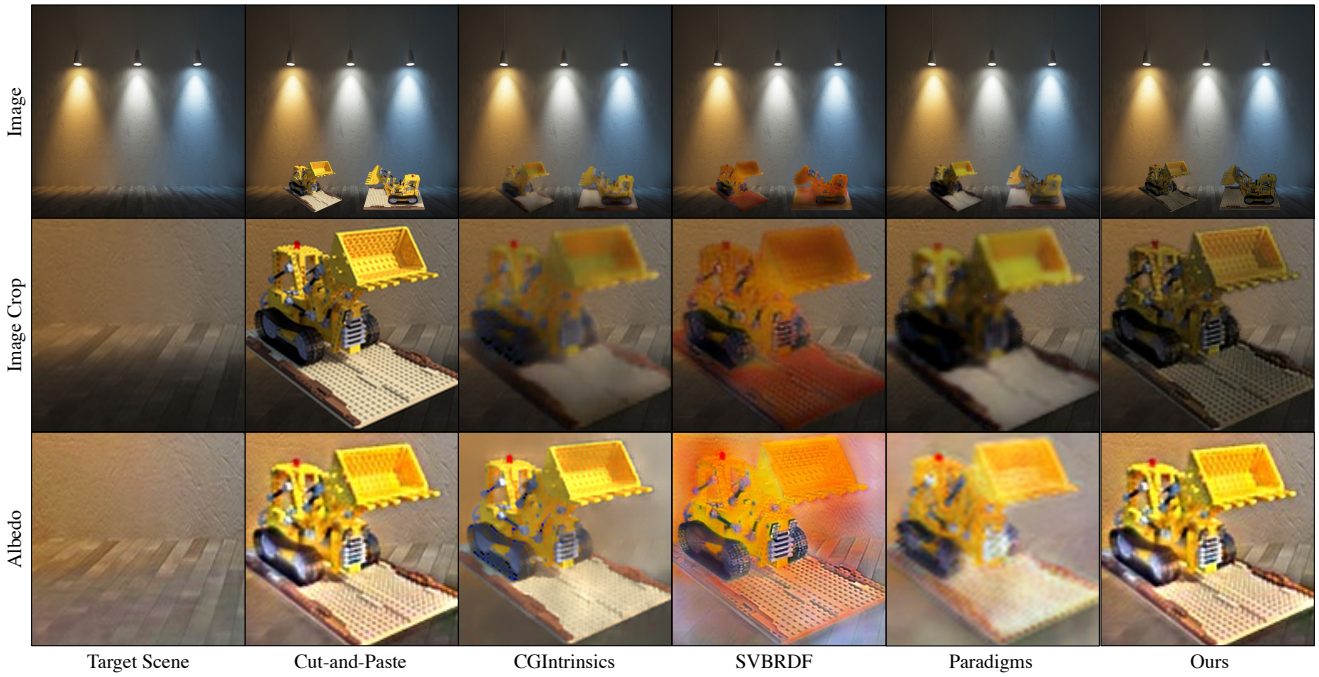
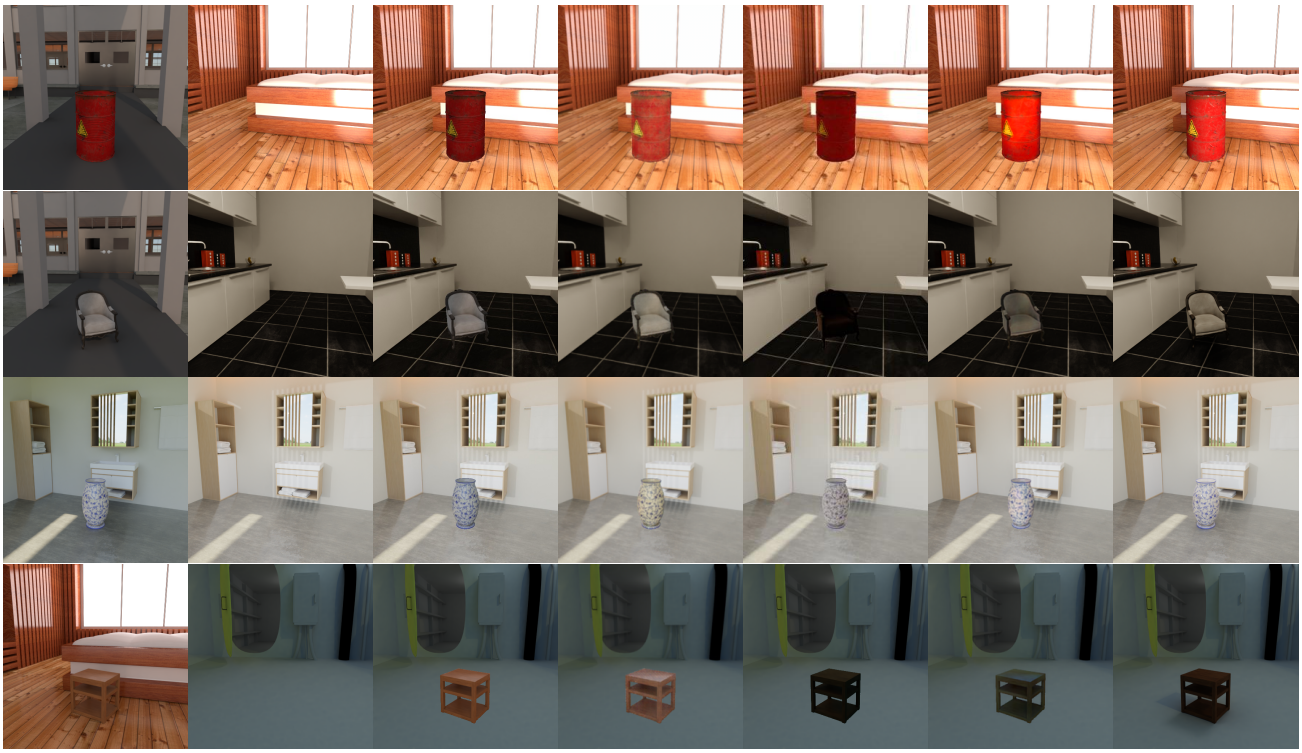


Figure 7. Current SOTA image decomposition methods do not recover geometric details when relighting objects. **Top:** relightings of a lego object inserted into the target scene using various methods. **Middle:** zoom on the front-end-loader. Cut-and-paste is too bright; other methods blur detail because high-frequency signals of the lego are modeled in the shading field. **Bottom:** shows albedo. Our modified decomposition (**right**) preserves surface details like studs on lego. These details may lead to albedo error but are crucial for insertion rendering to work. Note, the albedo for the target scene and cut-and-paste are using our modified decomposition.

inserted 2D image fragments without access to a surfaced 3D model of the fragment highlights the potential of learning abstract illumination representations.

Intrinsic images are seldom interesting in themselves; they are an interim representation used for some other purpose. It is likely better to evaluate decompositions in the context of their application, rather than for veridical accuracy. Doing so produces a predictable surprise: WHDR (a



Source | Target | Cut-and-paste | Dovenet [7] | Oracle [20] | Ours | GT

Figure 8. **Comparison on CG rendered dataset.** We compare our method (column 6) to a naive cut-and-paste baseline (column 3), a state-of-the-art image harmonization baseline (column 4), an upper baseline that uses GT 3D geometry (column 5), and ground truth images (last column). Illuminator can infer strong gloss correctly (red barrel in the top tow) that even a 3D supervised method cannot. It also generates plausible relighted rendering for complex materials like a ceramic vase (third row). We find the environment map predictions of the upper baseline [20] to be inconsistent in scenes with very low luminaire and strong background; thus resulting in incorrect rendering (see armchair). The images are chosen to show the distinct properties of Illuminator.

Figure 9. **Applications: scene relighting.** Illuminator can generate plausible relighted versions of the scenes by simply editing light sources. Our decomposition allows us to manipulate light sources either by adding or removing them in our gloss map or using transient latent variables from another image as in Fig. 3. For suppressing illumination from the light source on the left, we apply a smooth Gaussian kernel near the light source in the gloss map. For adding light sources, we randomly chose a gloss from our decomposition dataset (Fig. 4). We then use these updated gloss maps for relighting. On the left, notice glossy patches are removed on the floor and pillow. On the right, a realistic strong glossy reflectance on the metallic surface of the cellphone.

score of veridical accuracy) does not predict performance on a task-oriented metric.

Limitations: Illuminator currently cannot make cast shadows. However, methods that are trained to produce cast shadows from images [43, 30] (supervised) work only on outdoor scenes and only for very specific objects. We show some failure samples from [30] in our supplementary. Moreover, [43, 30] does not relight objects. But ideas from these methods may be complementary to Illuminator.

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Supplementary Material for Illuminator

A. In-The-Wild Object Insertion

We provide additional examples of in-the-wild object insertion in diverse spatially varying illuminated scenes. We insert a set of three buckets in Figure 10.

Target Scene Cut-and-Paste Illuminator

Figure 10. Additional examples for lighting correction of objects for in-the-wild images.

B. Synthetic evaluation dataset

We present a dataset constructed from 12 3D objects and 9 distinct background scenes. We illuminate each background scene with a small number of suitable illuminations, resulting in 23 target backgrounds, into which we aim to insert one of the 12 objects. By exhaustively matching objects and target backgrounds, we obtain 276 images, rendered for evaluation at 512 $\times$ 512 pixels and 500 samples per pixel with Blender's physically-based Cycle renderer. All images are post-processed by a denoise operation in Blender. The camera positions are fixed for the same object in all scenes so that we can easily relight the object in different background scenes.

Table 4 shows all objects used in our dataset. Most objects are collected from Poly Haven The Lego and materials are obtained from Blend Swap. We obtain the background scenes from TURBOSQUID.

Table 4. A complete list of twelve objects used in our dataset. The source for all these objects is hyper-referenced in the table for each object.

12 Objects		
Antique Ceramic Vase	Arm Chair	Barrel
Baseball	Lego	Marble Bust
Materials	Potted Plant	Side Table
Small Wooden Table	Vintage Cabinet	Wine Barrel

C. Additional experimental details

For our losses, we use $\lambda_{rec} = 10$, $\lambda_{per} = 0.01$, $\lambda_{decomp} = 10$ and $\lambda_{adv} = 1.0$.

Figure 11. An overview of all object in our dataset.

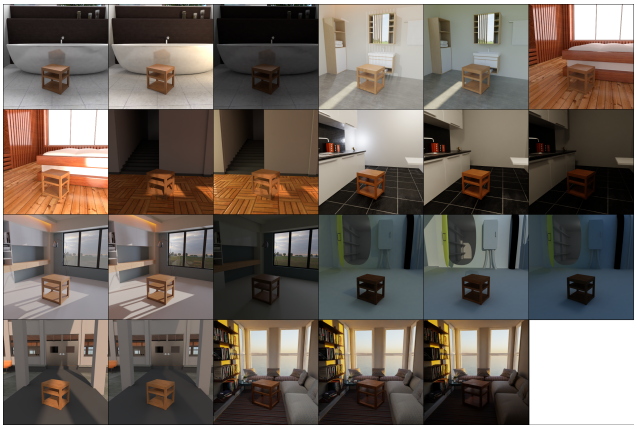

Figure 12. An overview of all 23 background scenes rendered with an exemplary side table from our set of objects.

Table 5. A complete list of scenes used in our dataset. The original source is hyper-referenced in the table for each scene.

9 Scenes		
Bathroom	Modern Bathroom	Wood Bedroom
Corridor	Kitchen	Living Room
Mech Room	Outdoor	Room

Our cross-loss also uses all the losses given in Equation **??** with the same lambda coefficients applied at different stages of abstracting and decoding of intermediate images and back to the original images. Except, our decomposition consistency loss during cross-loss is applied only to the shading for the intermediate reconstructed image.

Our persistent encoding, transient abstracting, and image decoding are simple U-Net with skip connections from the persistent encoder network to the image decoder. Our encoder has six down-blocks starting from 256×256 image resolution to 8×8 spatial feature code and six up-blocks to decode back an image with 256×256 resolution. Each down-block has two convolutions with a group normalization layer and then followed by a max-pooling layer. Each up-block has a bi-linear upsampling layer, followed by two convolutional with a group normalization layer. We use ReLU as our activation layer.

For our experiments, we use a batch size of 16 on a single A40 48GB GPU. We train illuminator with 10^{-4} learning rate and discriminator with 5×10^{-4}. We did not adopt any hyperparameter tuning and may expect better results with tuned hyperparameters.

D. Additional details on variant decomposition

Our variant albedo and shading decomposition are similar to paradigm reconstruction [10]. An encoder, with a ResNet structure produces an image code. This is decoded by a decoder (again with a ResNet structure) into an albedo map, a shading map, and a gloss map using three separate decoder heads. The albedo map is colored. Shading and gloss maps are colorless. The training process accepts albedo, shading, and gloss maps (generated as samples in advance and cached as shown in Figure 4) and real images. The real images are given without any of their ground truth (because we do not have one for MS-COCO images). We use real images to ensure our image decomposition inferences when recomposed back are able to produce an image that very much looks like real images. We update our decomposition network with this residual loss from real images after training with about 50k statistical samples of our persistent and transient maps. There are four losses to train our persistent and transient decomposition network. The decomposition network must decompose a fake image (constructed out of randomly selected persistent, multiplicative transient, and additive transient maps) into its correct, known decomposed maps (mixed L1/L2 loss). The persistent and transient maps constructed from a real image using our decomposition network must combine into that real image (mixed L1/L2 loss). The remaining two losses are adversarial. The persistent map constructed from a real image must fool a classifier trained to distinguish between fake persistent maps and those constructed from real images. Similarly, the transient maps constructed from a real image must fool a classifier trained to distinguish between fake transient maps and those constructed from real images.

Table 6. Ablation for choice of discriminator.

Discriminator	PSNR↑	LPIPS↓
Convolutional	17.42	0.038
Vision transformer	**17.65**	0.038

Fail 1 Fail 2 Fail 3

Figure 13. **Shadows from [30]** are unrealistic and not consistent with the shape of the inserted drum.

E. Additional quantitative results

We provide an ablation for performance for different objects in Table 7 and also an ablation for the choice of discriminator in Table 6.

F. Additional qualitative results

We show current SOTA shadow generation methods fails generating realistic shadows for indoor scenes in Figure 13. We tested our rendered objects with an off-the-shelf pretrained shadow generation network [30] (only publicly available code). We find the method struggles and produces unrealistic shadows for our objects and scenes. This is primarily because methods that are trained with only a few objects and outdoor scenes cannot generalize on any given indoor scenes for arbitrary objects. Outdoor scenes shadows have certain properties that are relatively easy to model [43]. However, indoor shadow generation is complex and a significantly hard problem. Even a minor error can cause unrealistic-looking images.

We also provide additional qualitative results comparing our renderings with baseline methods in Figure 14.

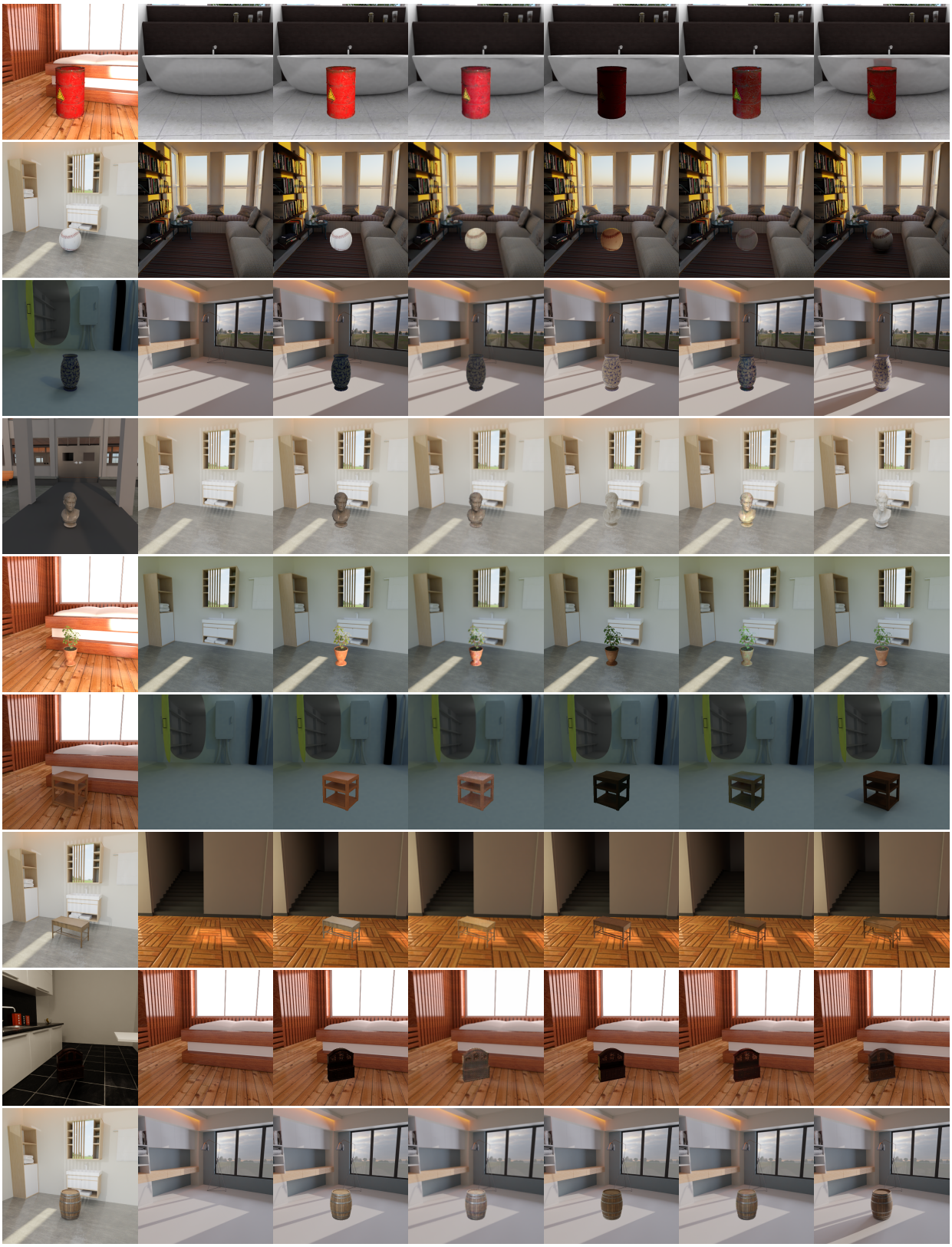

Figure 14. Additional random results comparing our method with an IH baseline, a 3D object supervised baseline and ground truth images. In each row, we show an object that is cut from a source scene and pasted into a target scene and renderings of different methods.

Table 7. We report PSNR scores for all objects used in our quantitative evaluation. Our decomposition with our cross-loss consistently outperforms for eight out of twelve objects we studied. We observe this strong performance as our method adds or removes gloss (Barrel in Figure 8) that is consistent with the background scene lighting. Baseline IH methods struggle to make these large appearance changes.

Object	IR++ [20]	CP	Dovenet [7]	Rainnet [28]	Ours baseline	Ours (w/o CICC)	Ours
Arm Chair	18.22	14.3	14.35	14.88	15.05	16.20	**16.76**
Barrel	21.46	17.16	16.42	16.71	18.82	19.59	**20.34**
Ceramic Vase	16.24	12.78	**15.64**	13.97	13.69	15.37	15.58
Baseball	14.95	12.18	**14.61**	13.82	12.99	14.55	14.46
Lego	18.45	14.56	16.18	15.76	15.65	16.30	**16.86**
Marble Bust	15.69	12.69	**15.56**	14.29	13.64	15.24	15.49
Materials	16.67	14.65	**17.81**	16.58	15.89	16.40	17.23
Potted Plant	19.39	15.32	17.60	16.97	16.43	17.58	**17.81**
Side Table	18.99	14.67	16.11	16.00	16.14	17.34	**17.69**
Wooden Table	21.02	16.69	18.48	17.89	18.25	18.96	**19.21**
Vintage Cabinet	24.26	19.58	18.94	20.06	20.81	19.64	**21.32**
Wine Barrel	20.72	16.42	17.63	17.83	17.54	18.75	**19.11**
Mean	18.84	15.08	16.61	16.23	16.24	17.16	**17.65**